



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-6: 2 Classes



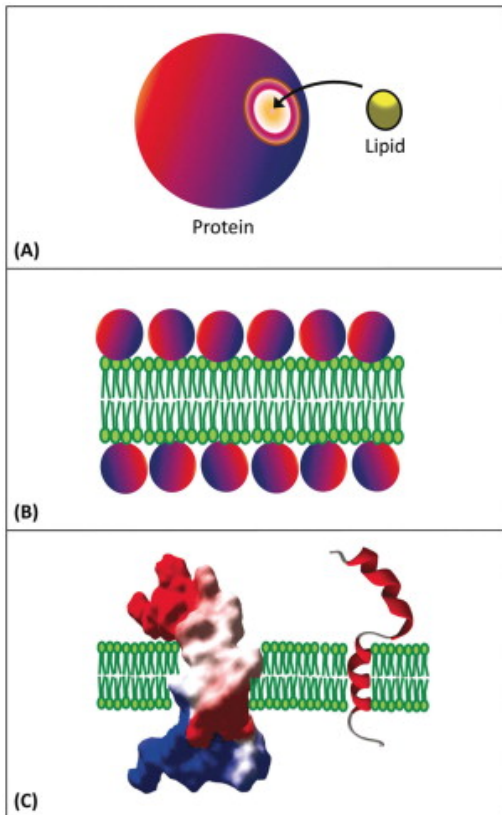
*Please Listen Carefully
&
Keep your phone(s) on silent mode*

Molecular Interactions

Structure-activity relationship

Molecular Interactions

- Molecular interactions are attractive or repulsive forces *between* molecules and between non-bonded atoms. These are important in all aspects of chemistry, biochemistry and biophysics, including protein folding, drug design, pathogen detection, material science, sensors, nanotechnology, separations, and origins of life.
- Molecular interactions - noncovalent interactions, intermolecular interactions, non-bonding interactions, noncovalent forces and intermolecular forces.



Evolution of the notion of protein–lipid interactions during the past 60 years.

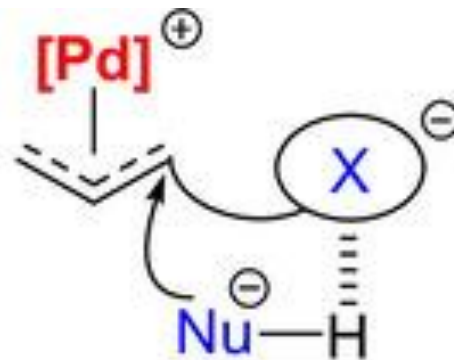
(A) Binding of a lipid molecule in a hydrophobic pocket of the protein. (B) Early model of the plasma membrane with proteins adsorbed on both sides and interacting with the head groups of phospholipids through electrostatic bonds. (C) Topology of an integral membrane protein inserted in the membrane bilayer (surface model on the left, ribbon model on the right). In this case, the protein may interact with membrane at three levels (extracellular, transmembrane, and intracellular domains).

Non-Bonding Interactions

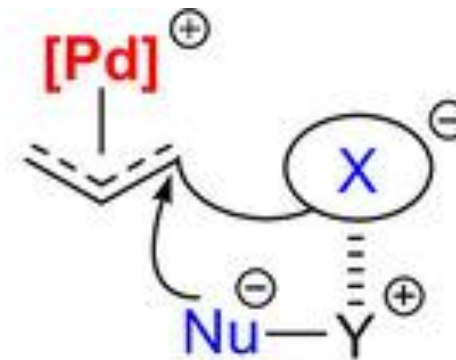
Non-Bonding Interactions. Molecular Interactions are between molecules, or between atoms that are not linked by bonds.

Molecular interactions include cohesive (attraction between like), adhesive (attraction between unlike) and repulsive forces between molecules. Molecular interactions change (and bonds remain intact) when (a) ice melts, (b) water boils, (c) carbon dioxide sublimates, (d) proteins unfold, (e) RNA unfolds, (f) DNA strands separate and (g) membranes disassemble.

The enthalpy of a given molecular interaction, between two non-bonded atoms, is 1 - 10 kcal/mole (4 - 42 kJ/mole), which in the lower limit is on the order of RT and in the upper limit is significantly less than a covalent bond.

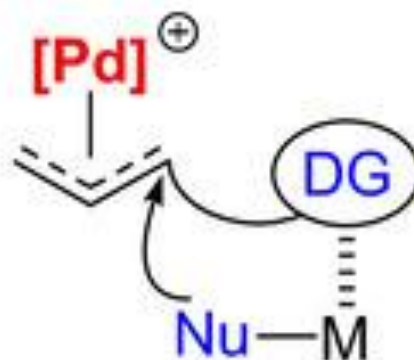


H-Bonding Interaction

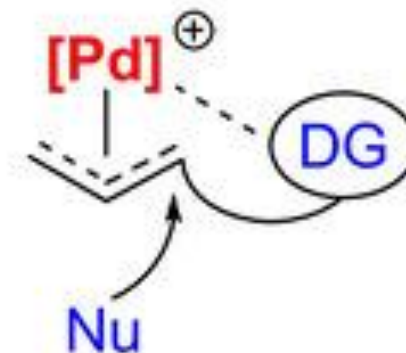


Electrostatic Interaction

Non-Bonding Interaction



or

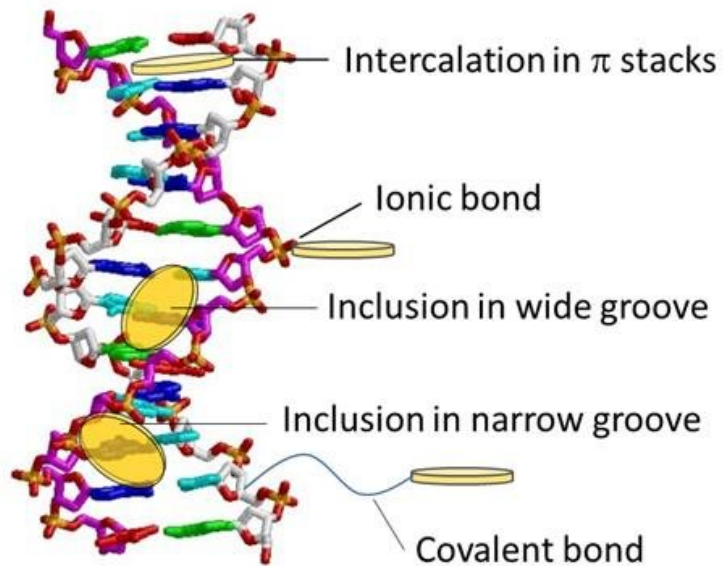


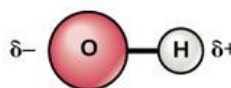
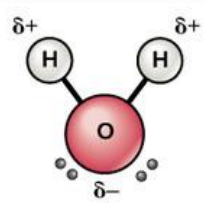
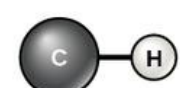
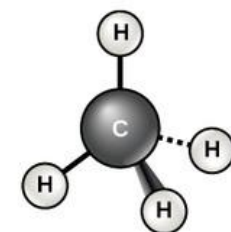
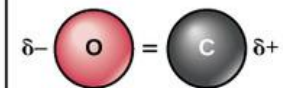
Coordination Interaction

Bonding interactions



- Bonds hold atoms together within molecules. A molecule is a group of atoms that associates strong enough that it does not dissociate or lose structure when it interacts with its environment.
- At room temperature, two nitrogen atoms can be bonded. Bonds break and form during chemical reactions. In the chemical reaction called fire, bonds of cellulose break while bonds of carbon dioxide and water form. Bond enthalpies are on the order of 100 kcal/mole (400 kJ/mole), which is much greater than RT at room temperature; bonds do not break at room temperature.

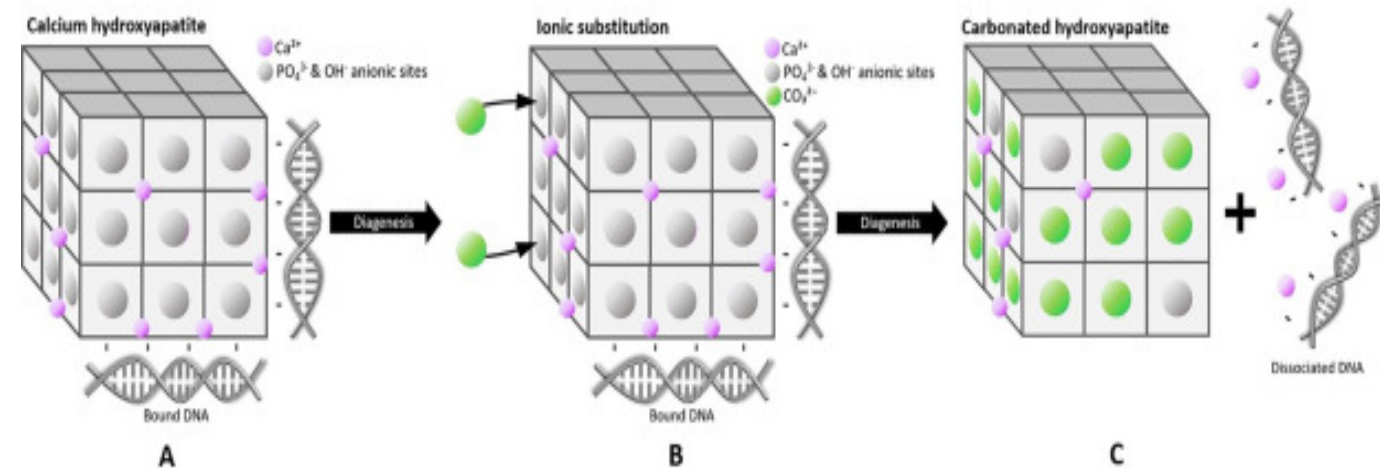


	Bond type	Molecular shape	Molecular type
Water	 Polar covalent	 Bent	Polar
Methane	 Nonpolar covalent	 Tetrahedral	Nonpolar
Carbon dioxide	 Polar covalent	 Linear	Nonpolar

Correlation between bone diagenesis and DNA preservation



- Molecular interactions between the inorganic hydroxyapatite matrix of bone and DNA have been strongly correlated to DNA preservation. Prior to diagenetic alteration, the negatively charged backbone of DNA molecularly interacts with positively charged calcium (Ca^{2+}) residues in hydroxyapatite.
- This association is supposed to protect DNA from physical and chemical damage. However, as the duration of environmental exposure increases, diagenesis of bone microstructure affects this affinity of DNA for the mineral matrix. As diagenesis progresses, ionic substitution and infilling displaces calcium ions and causes DNA to dissociate from the hydroxyapatite matrix. When this occurs, DNA becomes more susceptible to damage by environmental factors (e.g., heat, moisture, microbes, soil acidity).



Basic schematic of the mineralized hydroxyapatite matrix of bone, the interactions between DNA molecules and this microstructural component of bone, and the ionic substitutions that occur during diagenesis: (A) prior to diagenetic alteration, the negatively charged DNA backbone molecularly interacts with the positively charged calcium residues in the inorganic mineralized bone matrix, and this association has been shown to provide protection to DNA; (B) as diagenesis progresses, ionic substitution and infilling with carbonate and other ions occurs within the hydroxyapatite matrix; (C) as hydroxyapatite becomes more carbonated, calcium ions are displaced and leach out of the mineralized matrix, which causes DNA to dissociate and become more susceptible to damage by environmental factors.

CNTs interacting with brain cells



- CNTs are internalized by cells via endocytosis or a nano-penetration method. In the endocytosis, nano-scale CNTs are engulfed into endocytic vacuoles because of their small size and there they are destroyed by various mechanisms adopted by the cells such as oxidative stress or release of various chemical toxins.
- Endocytosis is an energy-dependent process which maybe hindered in low-temperature conditions as well as in ATP-deprived conditions. Functionalized CNTs wrapped with DNA or PEGylated are phagocytized to a greater extent by cells.
- The uptake mechanism and pathway of CNTs is via intracellular transporters for proteins and DNA. They observed that SWCNTs with various functionalization are capable of transporting protein and DNA by energy-dependent endocytosis into cells which occurs via clathrin-coated pits rather than

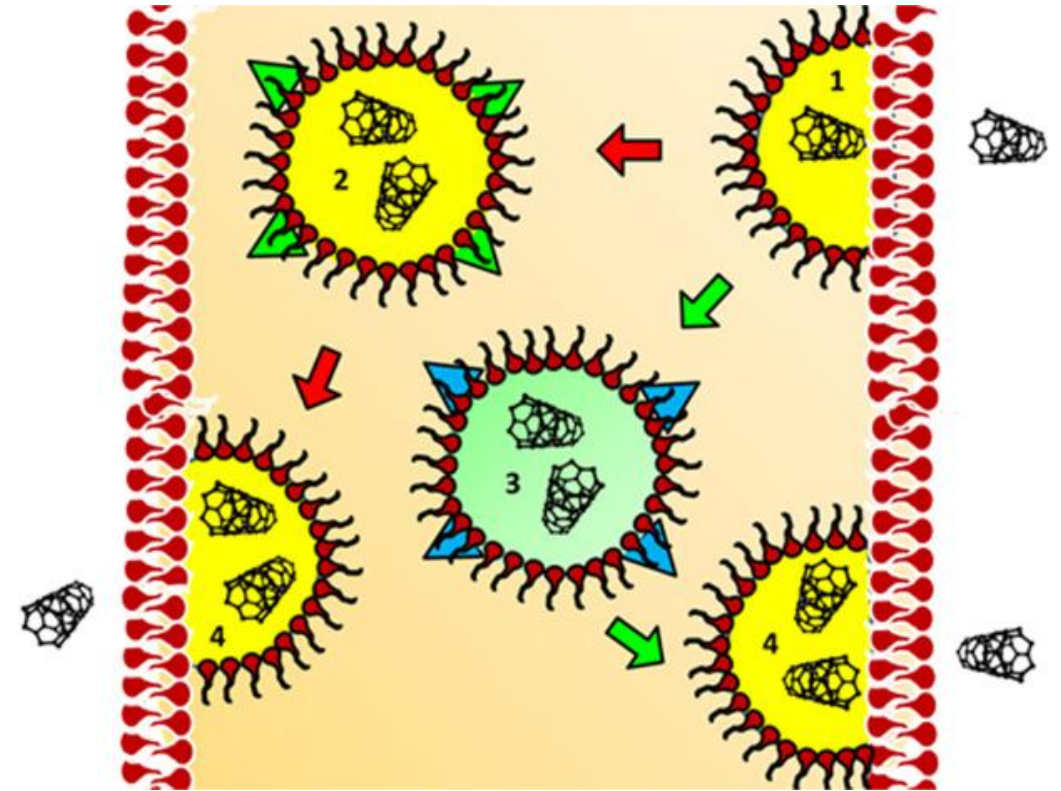
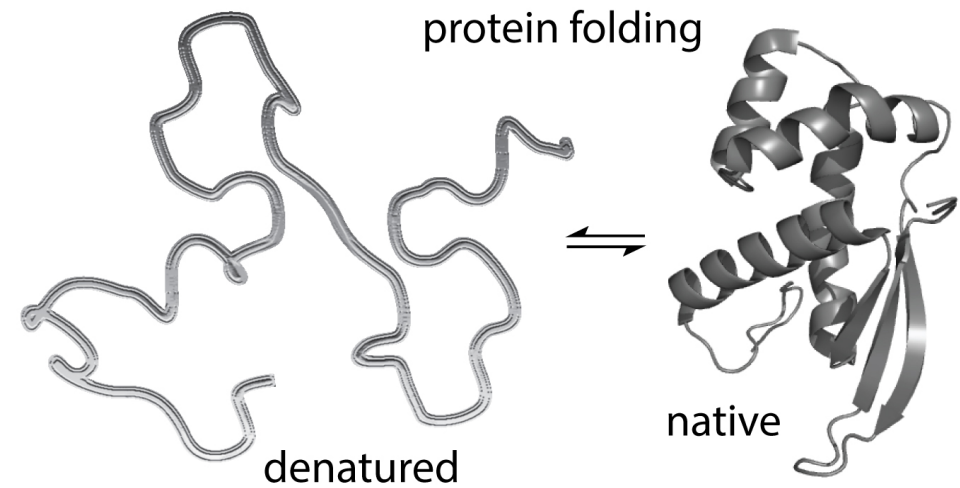


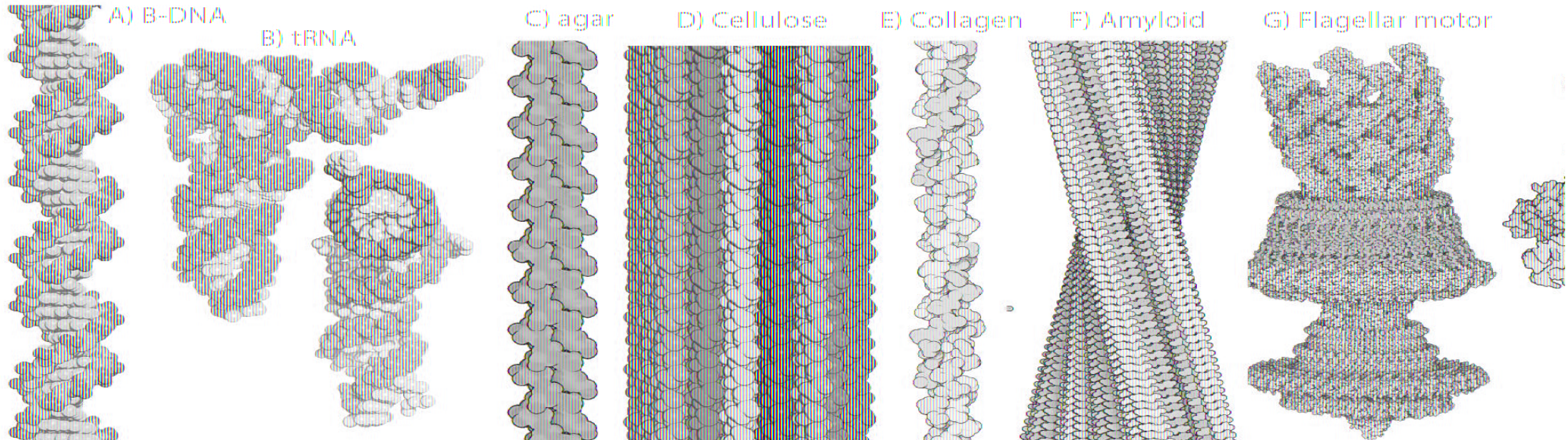
Illustration showing internalization of CNTs carrying drug molecules by endocytosis and also externalization from phagocytic cells by exocytosis. Both (1) endocytosis and (4) exocytosis processes are energy-dependent and are considered as major molecular mechanisms behind the uptake of CNTs into cells. The mechanism is further explained by (2) clathrin-coated vesicles as well as the (3) caveolae pathway.

Folding & Assembly of Biological Macromolecules



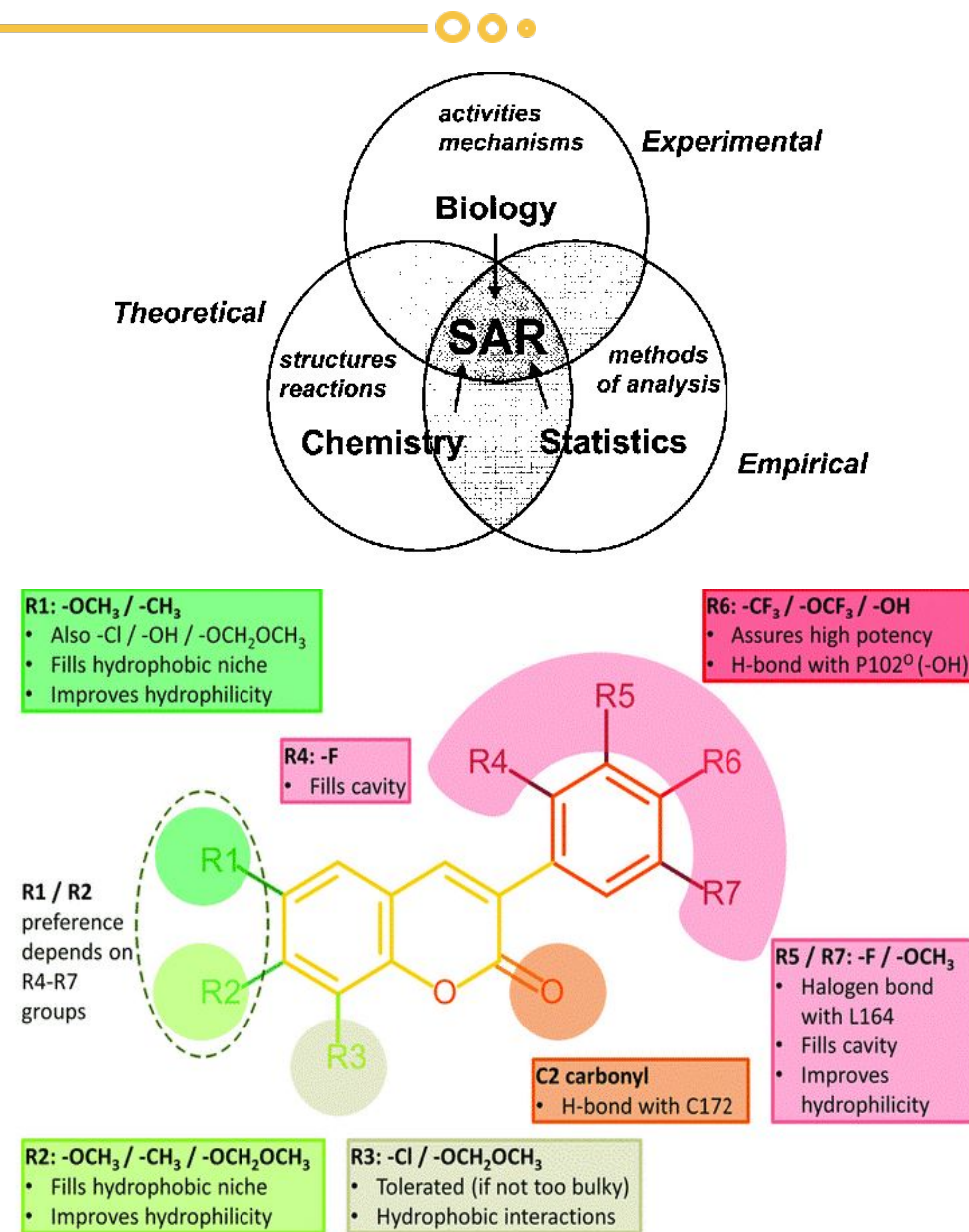
Protein folding is the conversion from a denatured state (a random coil) to a native state.

A) Two single strands of DNA assemble (anneal) to form a double helix; B) A transfer RNA (tRNA) molecule assembles (folds) to a stable L-shaped structure, C) Agar is a double-helical polysaccharide; D) Cellulose is an assembly of many polysaccharide strands; E) A collagen fiber is composed of three polypeptide strands; G) A flagellar motor is an assembly of five distinct types of folded polypeptide; H) The ribosome, a large assembly of around 50 polypeptide chains and over 3,000 deoxyribonucleotides.

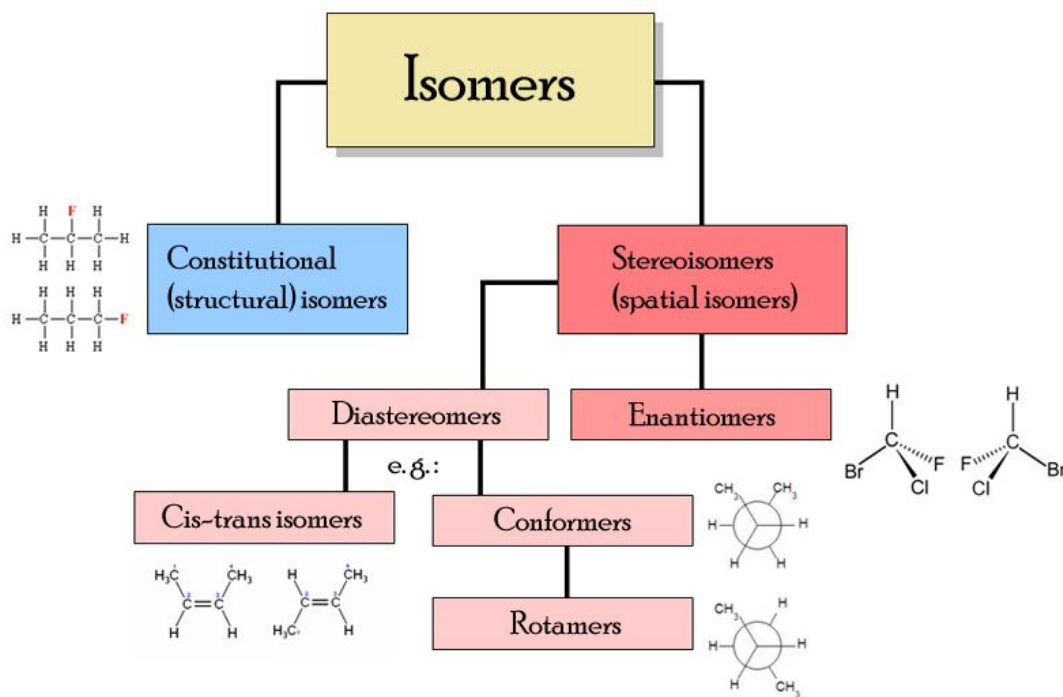


Structure-activity (SAR) relationship 5 marker

- The structure–activity relationship is the **relationship between the chemical structure** of a molecule and its **biological activity**.
- First presented by Crum-Brown and Fraser in 1865.
- Enables the determination of the chemical group responsible for evoking a target biological effect in the organism.
- Allows **modification of the effect or the potency of a bioactive compound** (typically a drug) by changing its chemical structure.
- To discover new drugs and also **repurpose existing drugs** – for the improvement in the biological activity.



Structure-activity (SAR) relationship

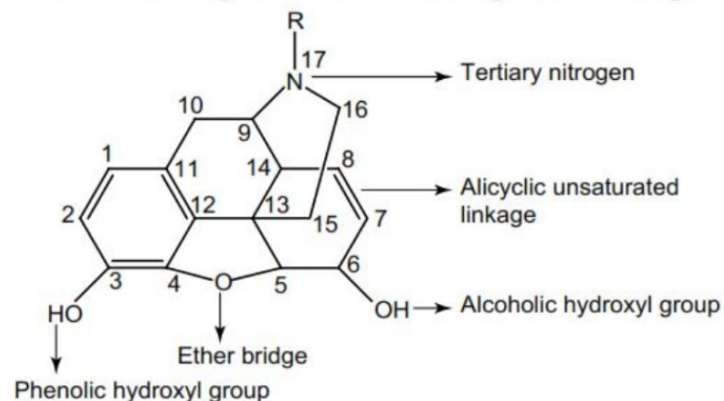


□ **Stereochemistry** involves the study of the relative spatial arrangement of atoms that form the structure of molecules and their manipulation.

□ It focuses on the relationships between stereoisomers and have the same molecular formula and sequence of bonded atoms but differ in the geometric positioning of the atoms in space.

□ A drug molecule can bind to a specific protein in the body and blocks its activity. This protein is involved in a disease process, and by inhibiting its activity, the drug can potentially treat the disease. To improve the drug's efficacy and reduce its side effects, chemical structure modifications can be performed and can be tested for their biological activity. **By systematically changing different parts of the molecule, they can identify which chemical groups are essential for the drug's binding to the protein and which ones can be modified without affecting its activity.** This process of modifying the structure and testing the activity is called

SAR of Morphine (Structure Activity Relationship of Morphine)





□ Why SAR exist?

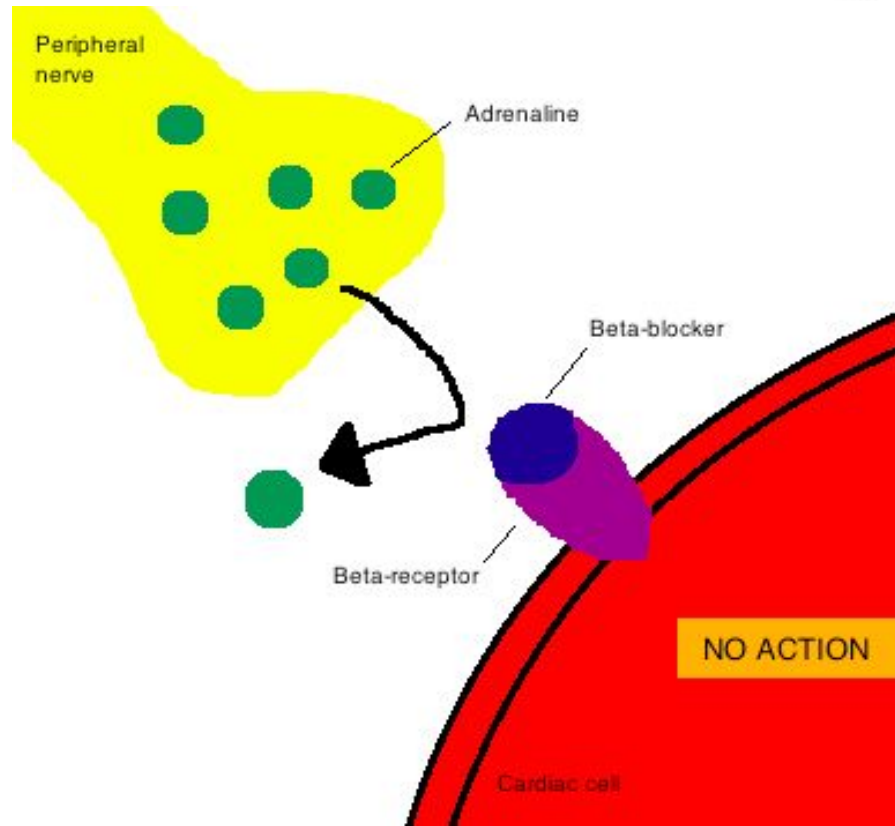
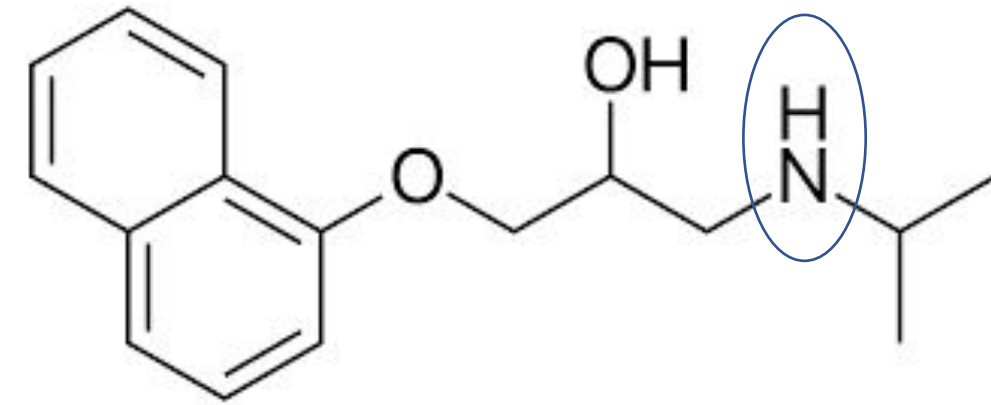
- To study the interaction of the drug molecule with protein and depends on its chemical structure.

□ What is the need for SAR study?

- A study of the SAR is mainly done by the lead molecule.
- It is used to determine the pharmacophore, unwanted side effects
- To develop a new drug that has better activity
- To determine some different activity from an existing drug
- To minimize side effects
- To know the chances in pharmacological properties by performing minor changes in the drug molecule

Examples of SAR – Beta blockers

Drugs used to treat hypertension, and other cardiovascular conditions. They block the activity of beta-adrenergic receptors in the heart, which reduces the heart rate and the force of the heart's contractions. The presence of tertiary amine is essential for their activity. Most of the beta blockers are in the chemical class of Aryloxypropanolamine.



Other examples of SAR

5 marker (SAR)

□ **Antibiotics:** **Penicillin** and its derivatives work by **inhibiting the enzymes involved in bacterial cell wall synthesis**, while **tetracyclines** work by **binding to the bacterial ribosome** and inhibiting protein synthesis.

□ **Antidepressants:** They work by **altering the levels of neurotransmitters** in the brain, such as **serotonin** and **norepinephrine**. The functional groups such as an **amine** or a **halogen** can affect their activity and selectivity for different neurotransmitter receptors. For example, selective **serotonin reuptake inhibitors** work by **blocking the reuptake of serotonin in the brain**, while selective norepinephrine reuptake inhibitors work

